Prem **Mallappa**

Systems Software Architect • Embedded Systems • Virtualization

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Seasoned Software Architect with extensive expertise in designing and developing high-performance embedded systems, system software, Linux kernel drivers, and virtualization technologies. Proficient in a wide range of programming and scripting languages, with a proven track record of delivering robust, scalable solutions in complex technical environments. A highly analytical and collaborative team player, recognized for strong problem-solving abilities, logical thinking, and a passion for mastering emerging technologies. Committed to driving innovation and excellence through clean architecture, optimized code, and best-in-class engineering practices.

Skills

Leadership	**22+ years** of multidisciplinary team and project leadership experience; **software architecture** and technical direction across systems and platform teams.
Communications	**Clear written and spoken communication** for architecture reviews, design discussions, mentoring, and cross-functional execution.
Technology	Broad and deep IT expertise across **embedded systems**, **system software**, **cloud/security**, **virtualization**, and software services development.
Computer Security	**Cryptography**, virtualization and cloud computing security architecture, **secure software development**, risk management, compliance, intrusion detection, and prevention.
Programming Languages	**C**, **C++**, Assembly (**x86**, **ARM**, **MIPS**), Rust, Python, Go, BASH
Operating Systems	**Linux**, FreeBSD, QNX, VxWorks, Symbian
Architecture	**x86**, **ARM v6/v7/v8**, **MIPS64**, PowerPC, RISC-V
Development Tools	**GCC**, **Clang**, **Git**, **CMake**, Make, GDB, Perf, VTune, ARM Development Tools, Keil, JTAG, Docker, QEMU
Systems & Virtualization	**Linux kernel development**, **device drivers**, hypervisors, **IOMMU/SMMU**, pre-silicon validation, boot flows, and low-level firmware bring-up.
Performance & Acceleration	**SIMD/vectorization**, **performance optimization**, math and crypto libraries, **SmartNIC/DPU offload**, NVMe-oF, RDMA, and hardware-assisted data paths.
Specialized Skills	Embedded systems, system software, **cryptography**, filesystems, performance engineering, packet/data-path acceleration, and platform enablement.

Education

Masters (Computer Science)
Pilani, Rajasthan, India

BITS Pilani
2016

Bachelors (Computer Science)
Chikkmagalur, Karnataka, India

Adichunchangiri Institute of Technology (AIT)
2002

Affiliated: Visvesvaraya Technological University (VTU)

Experience

AMD Ltd.

2018 – Present

PRINCIPAL ENGINEER

Worked in multiple teams within AMD including **AMD LibM**, **AOCL Cryptography**, **BSP**, and **NVMe** in **Pensando** BU. Every project was different, requiring quick learning and adaptation to new domains. The role also involved **leading a team** of four-to-six engineers in various projects. I was also part of the leadership team where I mentored engineers from across AOCL and Pensando BU.

SR. MEMBER OF TECHNICAL STAFF

Worked on **Model0** which was **pre-silicon validation** of next generation AMD cores, including Zen2 and Zen3. The role involved working with hardware team to bring up bare-minimum processor configuration with just core and memory controller, and run Linux, stressors and benchmarks. This involved working on low-level firmware, bootloaders, Linux kernel and performance analysis tools.

Broadcom Ltd.

2014 – 2016

PRINCIPAL ENGINEER

Architected foundational software stack for Broadcom Vulcan, a multicore-multithreaded ARMv8 64-bit server processor. Engineered complete **SMMUv3** (System Memory Management Unit) driver infrastructure supporting advanced IOMMU virtualization. Developed and contributed an **SMMUv3 emulation model** to QEMU mainline, enabling early platform validation and ecosystem enablement. Implemented comprehensive virtualization features including command queue processing, STE/CD descriptor parsing, multi-level page table walks, and stage-1/stage-2 address translation.

Cavium India Pvt. Ltd.

2011 – 2013

TECH LEAD

Led low-level software development for Cavium Octeon III (MIPS64) multicore network processors, delivering stable platform firmware for networking customers. Resolved a critical KEXEC kernel bug and authored the full **MIPS64 port** of the Kexec/Kdump crash dump infrastructure, ultimately merged into Linux mainline. Engineered a bare-metal core dump generation framework with a host-resident daemon that enabled detailed post-mortem GDB analysis during bring-up. Designed and implemented the **CavHv hypervisor** for MIPS64 architecture using hardware virtualization extensions to validate next-generation features.

ARM Ltd.

2005 – 2009

SR. DEVELOPMENT ENGINEER

Executed comprehensive **OS porting initiatives** for emerging ARM architectures including ARM1176JZFS with TrustZone security extensions and the Cortex-A8 application processor. Developed a production-grade touchscreen driver for Symbian OS along with a Python-based automated validation framework used across multiple programs. Designed a precision interrupt latency measurement infrastructure that quantified TrustZone secure-world context switching overhead for silicon partners. Ported the **L4Ka::Pistachio microkernel** to ARM architecture, enabling virtualization research and multiple proof-of-concept implementations.

Sasken Communication Technologies Ltd.

2004 – 2005

SR. SOFTWARE ENGINEER

Architected and maintained the **Extended File System (EFS)** for VxWorks-based UMTS base station infrastructure, balancing throughput with resiliency requirements. Engineered a reset-resilient filesystem with **advanced wear-leveling algorithms** so data integrity held across unexpected power cycles in the field. Designed a flash-optimized flat file tree structure that minimized write amplification and significantly extended device longevity.

Global Edge Software Ltd.

2003 – 2004

SOFTWARE ENGINEER

Designed and implemented a **comprehensive SDIO host controller driver** for Linux on Intel StrongARM embedded platforms used in consumer devices. Engineered a high-performance 4-bit SDIO driver for Marvell 802.11g WiFi chipsets that consistently achieved maximum theoretical throughput. Optimized **DMA transfers** and interrupt handling paths to minimize latency and maximize wireless data transfer efficiency under heavy load.

Short Stints

VSPL Ltd.

Software Architect

Nov. 2016 – Jan. 2018

Bengaluru, India

Cisco Ltd.

Software Engineer

Oct. 2010 – Apr. 2011

Bengaluru, India

B-Labs, London UK

Sr. Engineer - Contractor

Nov. 2009 – Sep. 2010

Bengaluru, India

Harman International

Engineer

Jul. 2009 – Nov. 2009

Bengaluru, India

Open Source

QEMU (Link)

2014-2016

- SMMUv3 (IOMMU) emulation support for ARMv8
- Designed and implemented SMMUv3 model merged into mainline
- Added support for Stage1, Stage2, and nested virtualization
- Implemented command queue processing and page table walk

Linux Kernel (Link)

2011-2016

- MIPS Kexec/Kdump port and IOMMU subsystem
- Developed MIPS64 port of Kexec and Kdump (merged upstream)
- Fixed critical bugs in KEXEC for Cavium Octeon platforms
- Contributed to IOMMU/SMMUv3 driver development

GLIBC (Link)

2018-2020

- Performance optimizations for AMD processors
- Fixed memcpy behaviour on AMD processors
- Optimized string functions for x86\\$_{64}$ architecture
- Performance improvements for memory operations

AMD LibM (Link)

2018-2025

- Open source math library for AMD processors
- Core contributor to open sourcing AMD Math Library
- Optimized transcendental functions (exp, log, pow, trig)
- Implemented SIMD/FMA optimizations for vector operations

Projects

AMD Ltd.

2018 – Present

PRINCIPAL ENGINEER

Bengaluru, India

DPU-Accelerated Virtio Live Migration

Jan. 2025 – Present

Synopsis Verification and addressing scalability issues on AMD/Pensando Elba ASICs during Live-Migration. Analyzed SEQ/ACK bitmap dirty-tracking protocol across VFIO and vDPA driver paths, identified vDPA implementation gaps, and built a full end-to-end suite covering NVMe+virtio-pci VM topology with automated QEMU orchestration and QMP-driven migration control.

Details Investigated **scalability bottlenecks** in the live-migration path on AMD/Pensando Elba ASICs by tracing dirty-page reporting from the **SEQ/ACK bitmap protocol** in the dataplane through firmware and host migration drivers. Compared **VFIO** and **vDPA** implementations, documented the missing dirty-tracking pieces in the vDPA path, and clarified where hardware capability was not being surfaced correctly in software. Built an **end-to-end validation framework** for NVMe + virtio-pci VM topologies with automated QEMU launch, disk-format detection, VF bind checks, hugepage preparation, and QMP-driven pre-copy / stop-and-copy orchestration. Used the framework to reproduce corner cases, validate migration sequencing, and establish a repeatable path for scaling analysis and future feature bring-up.

Languages: C, C++, P4, Python
Tools: Git, GDB, Qemu, Xen Hypervisor, DPDK
Tech: NVMe-oF, SmartNIC, DPU, Packet Processing Pipelines

AMD Pensando DSC Platform PCIe support

Jun. 2025 – Dec. 2025

Synopsis The DPU presents itself as a PCIe device, all access from the host ends up in indirect access and handled in software stack. Optimizting the PMT/PRT entries to make most of the hardware access end up as direct memory access.

Details Analyzed the host-to-DPU PCIe access path and identified where **indirect register and memory accesses** were introducing avoidable software overhead. Reworked **PMT/PRT programming** so frequently used host accesses could be resolved as **direct memory transactions** instead of trapping into slower software-managed flows. Validated the updated PCIe mapping behavior across host driver interactions and DPU-side handling to reduce access latency, simplify the datapath, and improve overall platform responsiveness for control-plane and bring-up operations.

Languages: C, C++, P4, Python
Tools: Git, GDB, Qemu, Xen Hypervisor
Tech: PCIe, SmartNIC, DPU, Hardware Offload

AMD Cryptography Library

Jan. 2023 – Jun. 2025

Synopsis Initiated as proof-of-concept to address ISV performance bottlenecks with AES-CFB cryptography on AMD EPYC Milan servers. Evolved into comprehensive cryptographic primitives library delivering production-grade implementations of AES and SHA algorithms optimized for AMD processor architectures.

Details Engineered parallel AES-CFB decryption algorithm delivering **3x performance improvement** over standard OpenSSL implementation. Developed multi-algorithm cryptographic library supporting AVX2, AESNI, and SHANI instruction set extensions. Implemented **CPUID-based runtime dispatcher** for automatic selection of optimal algorithm variant per processor generation. Integrated library as OpenSSL provider enabling transparent drop-in replacement for existing applications.

Languages: C++, x86-64 Assembly
Tools: Git, CMake, GCC, Clang, OpenSSL Provider API
Tech: AES-CFB, AES-GCM, SHA256, SHA512, AVX2, AVX-512, AESNI, SHANI, VAES

AMD Math Library (AOCL-libm)

Dec. 2019 – Dec. 2022

Synopsis High-performance mathematical library providing optimized implementations of transcendental functions (exp, log, pow, trigonometric) for AMD EPYC and Ryzen processors. Led comprehensive modernization initiative to enhance performance through advanced algorithmic techniques and microarchitecture-aware optimizations.

Details Implemented **Estrin's polynomial evaluation** method enabling parallel FMA instruction utilization for reduced latency. Developed vectorized lookup table architecture with cache-line alignment and interleaving for optimal memory bandwidth. Achieved **30% performance improvement** in scalar operations and **2x improvement** in vector operations across function portfolio. Engineered CPUID-based dynamic dispatcher selecting optimal code paths based on processor generation and feature flags.

Languages: C, x86-64 Assembly, Python
Tools: CMake, GCC, Clang, AOCC, Git, Perf, VTune
Tech: AVX2, AVX-512, FMA3, FMA4, Estrin Polynomial, SIMD Vectorization, Cache Optimization

SR. MEMBER OF TECHNICAL STAFF

Bengaluru, India

Model0

Feb. 2018 – Dec. 2019

Synopsis Next generation AMD processors needed to be tested before IO and other components are integrated, as part of the **shift-left** strategy. Allows faster integration and

Details As part of the Shift-Left strategy, full OS boot and CPU stressers were run early in the development before tape-out, with just the compute core and simulated memory, but no I/O. stressng and lmbench etc were run to stress the CPU code path. Efforts involvled trimming down Linux kernel enough so that it boots **without ACPI**, and in some cases with a very small ACPI table.

Languages: C, x86-64 Assembly, Python
Tools: CMake, GCC, Linux Kernel, Virtualization
Tech: AMD-SEV, AMD-SNP

Broadcom Ltd.

2014 – 2016

PRINCIPAL ENGINEER

Bengaluru, India

ARM IO Virtualization with SMMUv3

Jan. 2015 – Oct. 2016

Synopsis Developed comprehensive SMMUv3 (ARM System Memory Management Unit) emulation infrastructure enabling early software development and validation for Broadcom Vulcan ARMv8 server platform. SMMUv3 provides hardware-based IO virtualization with multi-stage address translation and device isolation capabilities.

Details Architected and implemented complete **SMMUv3 device model** for QEMU, successfully **upstreamed to mainline** repository. Developed command queue processing engine supporting asynchronous invalidation and event notification. Implemented ARMv8 LPAE page table walker supporting Stage-1, Stage-2, and nested translation modes. Designed comprehensive event reporting mechanism with configurable interrupt generation and fault handling.

Languages: C
Tools: QEMU Device Model, Git, GDB, ARM Fast Models
Tech: ARMv8, SMMUv3, IOMMU, 2-Stage Translation, Command Queue, Event Records

SMMUv3 Driver for Broadcom Vulcan

Jan. 2015 – Oct. 2016

Synopsis Broadcom Vulcan represents ground-up ARMv8 server processor design featuring 32 cores with simultaneous multithreading (4 threads per core). Engineered production Linux driver for integrated SMMUv3 enabling secure IO virtualization and DMA protection for datacenter workloads.

Details Developed initial **Linux kernel driver** implementing SMMUv3 core functionality including StreamID-based device isolation. Collaborated cross-functionally with hardware design team to identify and resolve silicon errata and performance bottlenecks. Contributed bug fixes and feature enhancements to ARM's reference SMMUv3 driver implementation. Designed and executed a **comprehensive validation suite** covering multi-stage translation and fault handling scenarios.

Languages: C
Tools: Git, GCC, ARM Cross Compiler, Kernel Build System
Tech: Linux Kernel Driver, IOMMU Framework, DMA API, Device Tree, MSI Interrupts

TECH LEAD

Bengaluru, India

Broadcom XLP (MIPS64) Platform Enablement

Jan. 2014 – Dec. 2015

Synopsis Broadcom XLP represents high-performance multicore MIPS64 processor architecture featuring up to 32 cores with 2-way multithreading and integrated hardware acceleration engines for networking and security workloads. Led Linux kernel enablement delivering production-grade device drivers and power management infrastructure.

Details Developed hardware random number generator driver providing high-entropy source for cryptographic applications. Implemented **Common Clock Framework** driver enabling dynamic frequency scaling across processor domains. Engineered DVFS (Dynamic Voltage and Frequency Scaling) driver reducing **power consumption by 40%** under variable workloads. Created CDE (Compression/Decompression Engine) driver offloading network packet processing to hardware accelerators.

Languages: C, MIPS Assembly
Tools: Git, GCC, MIPS Cross Compiler, Device Tree Compiler
Tech: MIPS64 Octeon, Common Clock Framework, DVFS, Hardware RNG, CDE Engine

Cavium Networks

2011 – 2013

TECH LEAD

Bengaluru, India

Cavium OCTEON III - Kexec/Kdump Implementation

May. 2011 – Dec. 2013

Synopsis Kexec enables fast kernel reboot without firmware re-initialization, while Kdump leverages this mechanism for capturing crash dumps. Developed complete MIPS64 architecture port enabling zero-downtime kernel updates and comprehensive crash analysis for Cavium OCTEON network processors.

Details Architected and implemented complete **MIPS64 port** of Kexec/Kdump infrastructure supporting OCTEON's Hybrid SMP topology. Resolved critical architectural bug preventing Kexec functionality on MIPS platforms. Successfully **upstreamed two kernel patches** to Linux mainline, now universally deployed across MIPS64 systems. Enabled seamless kernel hot-patching without disrupting Hybrid SMP core configurations or network traffic.

Languages: C, MIPS64 Assembly
Tools: Git, GDB, Crash Analysis, Kernel Build System
Tech: Kexec, Kdump, ELF Boot Protocol, Hybrid SMP, Memory Reservation, Crash Kernel

CavHv Hypervisor for MIPS64

Jan. 2012 – Dec. 2013

Synopsis Experimental Type-1 hypervisor leveraging MIPS Virtualization (VZ) hardware extensions for Cavium OCTEON-III processors. Collaborated with MIPS architecture team and Cavium hardware designers to validate draft virtualization specification through practical implementation.

Details Designed and implemented **bare-metal Type-1 hypervisor** based on MIPS Virtualization Extensions draft specification. Developed guest context management supporting CP0 register virtualization and root/guest mode transitions. Successfully demonstrated **concurrent execution of two independent Linux kernel instances** with isolated address spaces. Provided critical architectural feedback influencing finalization of MIPS VZ specification.

Languages: C, MIPS64 Assembly
Tools: GCC, MIPS Toolchain, GDB, QEMU
Tech: MIPS VZ Extensions, Type-1 Hypervisor, CP0 Virtualization, Root/Guest Mode

Cisco Systems

2010 – 2011

SOFTWARE ENGINEER

Bengaluru, India

Memory Leak Detection for Cisco IOS

Oct. 2010 – Apr. 2011

Synopsis Cisco IOS operates in both bare-metal and Linux environments with multi-month uptimes, making memory leak detection critical. Developed non-intrusive garbage detection system enabling proactive identification of unreachable memory in long-running network infrastructure.

Details Architected **custom C runtime wrapper** intercepting malloc/free operations without application source modifications. Implemented concurrent monitoring thread executing intrusive mark-and-sweep algorithm during idle periods. Designed reporting mechanism distinguishing legitimate long-lived allocations from leaked memory. Enabled field deployment teams to diagnose memory leaks before service-impacting failures occurred.

Languages: C
Tools: GCC, Glibc Hooks, Clearcase, Runtime Analysis
Tech: Mark-Sweep Algorithm, Malloc Interposition, Conservative GC, Graph Traversal

B-Labs

2009 – 2010

SR. ENGINEER - CONTRACTOR

London, UK (Remote)

CodeZero Hypervisor ARM Port

Nov. 2009 – Sep. 2010

Synopsis CodeZero represents microkernel-based hypervisor architecture targeting ARM Cortex-A9 multicore processors. Executed comprehensive platform bring-up enabling bare-metal execution on emerging ARMv7 SMP hardware.

Details Ported complete CodeZero hypervisor infrastructure to ARM Cortex-A9 SMP platforms including cache coherency management. CodeZero follows the **L4 microkernel family** model with a minimal kernel focused on address spaces, threads, IPC, and capability-style isolation, which shaped the platform bring-up and guest integration work. Developed **ARM-optimized memcpy** implementation leveraging NEON SIMD instructions for enhanced performance. Extended QEMU device models supporting CodeZero-specific platform requirements. Ported Linux kernel as guest OS executing under CodeZero hypervisor supervision.

Languages: C, ARMv7 Assembly
Tools: Git, QEMU, ARM GCC, JTAG Debugger
Tech: ARM Cortex-A9 MPCore, SMP, Cache Coherency, NEON SIMD, Microkernel IPC

Harman International

2009 – 2009

ENGINEER

Bengaluru, India

QNX BSP for Freescale PowerPC Platforms

Jul. 2009 – Nov. 2009

Synopsis Board Support Package development for QNX real-time operating system on Freescale QorIQ P4080 (8-core) and P2020 (dual-core) PowerPC processors targeting automotive infotainment systems. Enabled QNX deployment on next-generation multicore embedded platforms.

Details Resolved critical **SMP initialization bug** preventing P4080 8-core bring-up on QNX RTOS. Ported I2C controller driver enabling communication with touchscreen, audio codecs, and sensor peripherals. Developed foundational BSP infrastructure including timer subsystem and interrupt controller drivers. Successfully achieved **early boot milestones** on both P2020 and P4080 reference platforms.

Languages: C, PowerPC e500 Assembly
Tools: QNX Momentics IDE, Make, PowerPC Cross Compiler
Tech: QorIQ P4080, P2020, e500mc Core, AltiVec, SMP Spinlocks, I2C Protocol

ARM Ltd.

2005 – 2009

SR. DEVELOPMENT ENGINEER

Bengaluru, India

Multi-OS Enablement for ARM Processor Portfolio

Aug. 2005 – Jun. 2009

Synopsis Platform engineering team responsible for enabling diverse operating systems on emerging ARM processor cores. Supported major OS ecosystems including Linux, Symbian, and QNX across ARM11 (ARM1176JZF-S with TrustZone) and Cortex-A8 architectures, facilitating early partner evaluation and ecosystem development.

Details Developed **production-grade resistive touchscreen driver** for Symbian OS with multi-touch gesture recognition. Architected **Python-based test automation framework** reducing Symbian OS validation cycle time by 60%. Designed precision interrupt latency measurement infrastructure quantifying TrustZone secure-world context switch overhead. Ported L4Ka::Pistachio microkernel to ARM architecture enabling virtualization research and proof-of-concepts.

Languages: C, ARMv6/v7 Assembly, Python
Tools: ARM Fast Models, DS-5 Debugger, RealView Compiler, SoCDesigner
Tech: ARM1176JZF-S, Cortex-A8, TrustZone, Symbian OS, L4 Microkernel, VFP

Sasken Ltd.

2004 – 2005

SR. SOFTWARE ENGINEER

Bengaluru, India

EFFS - Embedded Flash Filesystem for UMTS Infrastructure

Aug. 2004 – Aug. 2005

Synopsis Designed and maintained Extended Flash File System (EFFS) for VxWorks-based UMTS base station controllers. Addressed unique constraints of Flash storage including limited erase cycles, asymmetric read/write performance, and power-failure resilience requirements for telecom infrastructure.

Details Architected **power-failure-resilient filesystem** ensuring data integrity across unexpected resets in field deployments. Developed **wear-leveling algorithm** distributing write operations across Flash blocks extending device lifespan by 3x. Implemented atomic transaction mechanism with flat file tree optimized for Flash characteristics. Integrated filesystem with VxWorks I/O subsystem maintaining POSIX-like API compatibility.

Languages: C
Tools: Tornado IDE, Wind River Compiler, VxWorks RTOS
Tech: NOR Flash, Wear Leveling, Erase Block Management, Power-Fail Safe, POSIX I/O

Global Edge Software

2003 – 2004

SOFTWARE ENGINEER

Bengaluru, India

Linux SDIO Host Controller Driver for Wireless Connectivity

Jun. 2003 – Aug. 2004

Synopsis Developed high-performance SDIO host controller driver for Intel StrongARM embedded platforms enabling wireless connectivity through Marvell 802.11g WiFi chipsets. Optimized for maximum throughput on resource-constrained ARM processors targeting portable and embedded applications.

Details Architected and implemented complete **SDIO host controller driver** supporting 4-bit wide bus mode and clock management. Engineered **DMA-based transfer mechanism** achieving maximum theoretical 802.11g throughput (54 Mbps) on StrongARM platform. Optimized interrupt handling and data transfer pathways minimizing CPU overhead and latency. Developed comprehensive integration with Marvell WiFi chipset firmware achieving production-grade wireless connectivity.

Languages: C
Tools: Git, GCC, GDB, ARM Cross Compiler, Logic Analyzer
Tech: Intel StrongARM SA-1110, SDIO Protocol, 802.11g WiFi, DMA Engine, IRQ Handling

Awards

Q3 2018 **Spotlight** *AMD*

Helping Emulation team find design bug in data fabric

Q2 2019 **Spotlight** *AMD*

Optimization of exp() and improving performance by 30%

Q2 2019 **Director Spotlight** *AMD*

Supporting Arden Ubuntu emulation boot, fixing initramfs delayed mount

Q4 2020 **Spotlight** *AMD*

Providing important patches to GLIBC to fix memcpy behaviour on AMD, open sourcing AMD LibM

Q3 2021 **Spotlight** *AMD*

Delivering Cryptography PoC (CFB based parallel decrypting), increases performance by 3x

2016 **Recognition** *Broadcom*

Upstreaming SMMUv3 emulation and delivering Linux driver infrastructure that enabled early ARMv8 server platform bring-up and validation

2013 **Recognition** *Cavium*

Delivering the MIPS64 Kexec/Kdump port and virtualization bring-up work that improved Octeon III platform readiness and post-mortem debug capability

2008 **Recognition** *ARM*

Driving ARM platform enablement through OS porting, automated validation, and L4Ka::Pistachio microkernel work that expanded early ecosystem readiness